TRIVIJAYA QUANT RESEARCH PROGRAMME

Paper 2 of 3 · Measurement infrastructure for AI-driven quantitative research on Indian equities

RegimeStress

A strategy survived one history.
When does it break?

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A benchmark for measuring how much a strategy's performance moves when the market environment is varied — rather than how high it is on the one environment that happened to occur.

Code, corpus, fixtures and evaluation protocol:
github.com/Ashwin-aadi/Trivijaya-Quant-Research
Programme overview and results: ashwin-aadi.github.io/Trivijaya-Quant-Research

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Abstract

A backtest is one draw from a distribution of histories that never got run. Indian equity markets over our study window contain roughly one pandemic crash, one full rate cycle and one structural liquidity shift, so a claim that a strategy is "robust across regimes" rests on a handful of effective observations no matter how many sessions were simulated. Paper 1 of this programme asked whether a machine-written strategy is real. The survivors it handed forward survived *one sequence of events*, and that is a weaker claim than it looks.

We release RegimeStress: a benchmark, protocol and reference implementation for measuring how much a strategy's performance moves when the environment changes. It provides a causally-labelled regime series over 1,233 sessions, estimated by expanding-window refits with a forward-only decode so that no label uses information from after the session it describes; **Counterfactual Regime Resampling**, a stationary block bootstrap conditioned on regime label that manufactures plausible histories that did not happen; a moment-validation suite those synthetic histories must pass; two stress tiers of very different cost; and a frozen strategy population whose exclusion rules were fixed *before* any stress path was run.

The central methodological result is that a shortcut reproduced performance and did not reproduce fragility. Resampling a strategy's realised returns agrees with full counterfactual re-simulation at Spearman 0.897 on mean performance but only 0.620 on fragility ($n = 125$) — and fragility is the quantity of interest. A second shortcut, measured the same way, holds at 0.963 and is adopted. **Shortcut safety is an experiment here, not a design decision.**

The benchmark's first prediction task is a null, and we diagnose it rather than report it bare. Fragility cannot be predicted from strategy characteristics at this sample size: five models on identical folds reach an out-of-sample R^2 of +0.024 across 109 strategies and 26 features. A permutation test nonetheless rejects the null for the *ranking* while failing to reject it for the *level*. Trained unchanged and applied to 60 strategies from three frontier models as held-out data, the predictor reaches $R^2 = -0.004$: no better than a constant. **The failure is a property of the problem, not of one small model's output.**

Fragility itself proved generator-independent — every frontier arm's median lands within a sixth of the local corpus's — while one measured pathology did not recur at all. We report our own two corrections in the results, because the first version of our headline number was better and wrong.

Keywords: regime switching • block bootstrap • strategy fragility • counterfactual resampling • hidden Markov models • negative results • Indian equities

Key findings

Each number answers a question, carries its sample size, and is tagged with the population it describes.

1. Can we test a strategy against histories that never happened? [GENERATION METHOD]

Yes, and the synthetic histories pass their own validation. 100 synthetic market histories, 124.5 CPU-hours of counterfactual backtesting. Of 9 target moments the real series must plausibly belong to, the conditional resampler fails **0** and the unconditional one fails 1. Conditioning on regime is what makes the synthetic histories admissible.

2. Can we do it cheaply instead? [GENERATION METHOD]

Not for the quantity we care about. The cheap tier reproduces the expensive tier at Spearman 0.897 on *mean performance* and only 0.620 on *fragility* ($n = 125$). **A shortcut reproduced performance. It did not reproduce fragility.** The expensive tier is not optional, and we know that because we ran both.

3. Can fragility be predicted without running the test? [GENERATOR-INDEPENDENT]

No. Out-of-sample R^2 =+0.024 on the primary target across 109 strategies. Applied to 60 frontier strategies as held-out data it reaches -0.004 — indistinguishable from predicting the mean. The limitation is *variance, not bias*: every model fits its training folds far better than it generalises, and the gap is widest for the model with the most capacity. So we added no capacity.

4. Does fragility depend on who wrote the strategy? [GENERATOR-INDEPENDENT]

No, and that is the result the benchmark most needed. Median fragility across regimes is 0.618 on the local corpus ($n = 125$) against 0.566, 0.624 and 0.696 on three frontier arms ($n = 20$ each). Every arm lands within a sixth of the local median. **What this benchmark measures survives a change of generator.**

5. Is a low fragility score good news?

Not by itself, and this is not a rare edge case. 16 of 125 strategies have a negative Sharpe in *every* labelled regime. The most extreme ranks 5 of 125 on the low-fragility end — among the most robust strategies we measured — and it never makes money. **A fragility score is uninterpretable without a performance figure beside it.**

6. Our own first answer was better, and wrong. [EXPLORATORY]

Deduplicating the corpus revealed 11 clusters of strategies with identical realised return series. A duplicate sitting in a training fold with its twin in the test fold was worth **0.238 of R^2** — the difference between a modest positive result and nothing at all. A second defect had one feature column duplicating another under a different name. Both corrections made the result worse, and both are in the results rather than a footnote.

One programme, three questions. Each paper answers the question the previous one raises, on one shared universe, one shared cost model and one shared backtest engine.

1	**AlphaAudit**	Can an AI produce a trading strategy that is real, rather than a plausible artefact of luck and leakage?
2	**RegimeStress**	If a strategy looks real, when does it break?
3	**FlowState**	If a strategy survives, how much capital can it absorb?

Code → validity → robustness → scale. The programme's overarching question is whether making the AI better makes the *research* better — and the three papers answer it only to the extent their experiments support.

Contents

1 The question

Paper 1 asked whether a machine-written strategy is real, and for most of a 14.6%-executable corpus the answer was no. The survivors — low-turnover strategies that cleared static leakage screening, a semantic consistency check, transaction costs, and one never-touched holdout year — are the input to this paper.

That is a weaker claim than it sounds, and the weakness is not statistical in the usual sense.

> A strategy backtested over five years of Indian equities has been tested against **one sequence of events**. Not five years of independent evidence — one draw. The window contains approximately one pandemic crash, one full rate cycle and one structural liquidity shift. **The number of independent regime observations is very small, and no amount of daily resolution increases it.**

Simulating 1,232 sessions does not give you 1,232 independent tests of whether a strategy survives a liquidity drought. It gives you one liquidity drought, sampled finely. A strategy that made money over that window might be robust, or might be well-suited to the particular ordering of events that occurred. Those two possibilities look identical in a single backtest, and distinguishing them is what this paper is for.

1.1 What "fragility" means here, precisely

For a strategy s evaluated across a set of market environments indexed by r, define

$$F(s) \;=\; \frac{\mathrm{sd}_r[\pi_r(s)]}{|\mathrm{mean}_r[\pi_r(s)]|},$$

where $\pi_r(s)$ is the strategy's performance in environment r. It is a coefficient of variation: how much the answer moves relative to how big the answer is. Low fragility means the strategy performs similarly wherever you put it.

Two things follow immediately, and both matter later. **Fragility scores consistency, not quality** — a strategy that loses money identically everywhere scores as admirably stable (§5.3). And it has a mean in its denominator, so a strategy whose average performance is near zero has an inflated score for purely arithmetic reasons.

1.2 What this paper does and does not claim

1. **It builds histories that did not happen**, and validates them against the one that did. §3.3.
2. **It measures which computational shortcuts are safe**, by running the expensive version and comparing. §4.
3. **It fails to predict fragility from strategy characteristics**, and diagnoses why. §6.
4. **It tests whether any of this depends on the generator**, using three frontier models and 6 generation methods. §9.

It does *not* claim that its synthetic histories are what would have happened. They are draws from a resampling scheme with stated assumptions, and §12 treats that as the paper's load-bearing limitation rather than a caveat.

2 What was available, and why it was insufficient

Regime-switching models (Hamilton, 1989; Ang and Timmermann, 2012) give us the labelling machinery and are not our contribution. What the literature typically does with them is fit on the whole sample and then evaluate strategies within the resulting labels — which uses information from after each session to describe that session. That is the same leakage paper 1 was built to detect, and §3.2 reports what it costs to avoid it.

Block bootstrap methods (Efron, 1979; Politis and Romano, 1994; Politis and White, 2004) give us the resampler. What they do not supply is a reason to believe a resampled path is a plausible *market*, as opposed to a statistically valid shuffle. §3.4 is the validation step that turns one into the other, and it is where the unconditional resampler fails.

Machine learning for asset pricing (Gu et al., 2020) predicts returns from characteristics. Predicting *fragility* from characteristics is a different and cheaper-to-label problem, since the target can be computed rather than waited for. It is also, on our sample, not solvable (§6).

> The gap RegimeStress fills: there was no released environment in which strategy fragility could be measured against validated counterfactual histories, on a frozen population with exclusion rules fixed in advance, with the expensive and cheap measurement paths both run so that the difference between them is a number rather than an assumption.

3 The benchmark

3.1 The population, and exclusions frozen before any path was run

The strategy population is paper 1's audit survivors plus a set of standard academic factors as positive controls. **Every exclusion rule below was fixed before a single stress path was run**, which is the only way an exclusion rule can be honest: applied afterwards, it is indistinguishable from dropping the inconvenient cases.

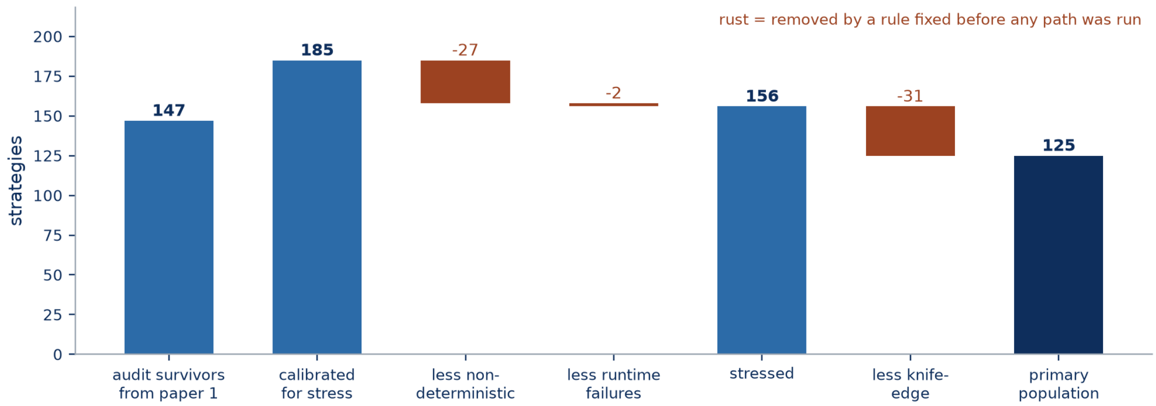

Figure 1. From paper 1's audit survivors to the population this paper measures. The rise at the second step is the standard academic factors joining as positive controls. Each rust bar is an exclusion, each exclusion is non-random, and each is reported as a limitation in §12. **Every rule that produced a rust bar was fixed before a single stress path was run**, which is the only way an exclusion rule can be honest: applied afterwards, it is indistinguishable from dropping the inconvenient cases.

3.2 Causal regime labelling

A hidden Markov model (Rabiner, 1989) over realised volatility, liquidity and macro state labels each session with one of K regimes. $K = 4$ was selected by BIC (Schwarz, 1978) over 1,207 sessions ending 2019-12-31 — **by information criterion, not by which labelling looked most recognisable**.

The labelling is **causal**. The model is refit on an expanding window (20 refits, from 1,207 training sessions to 2,378) and decoded forward-only, so the label attached to a session uses no information from after it. A full-sample fit would have been easier and is what the literature usually does.

What causality costs, reported rather than smoothed

21.8% of causal labels disagree with what the same estimator says at the end of the sample, and the labelling revises itself at a rate of 9.5% as data arrives. That drift is not a defect to be tuned away — it is what it looks like to label regimes without hindsight, and a paper that reports clean stable labels has probably used the future to get them. Regime occupancy ranges from 237 to 362 sessions across the 4 states, so no state is a degenerate corner.

A hand-curated timeline of 18 dated SEBI structural events ships beside the labels, each with its source, so that a reader can check whether a labelled transition corresponds to something that actually happened.

3.3 Counterfactual Regime Resampling

History supplied one path. **CRR manufactures more of them.**

The mechanism is a stationary block bootstrap (Politis and Romano, 1994) over *dates*, conditioned on regime label. Resampling dates rather than individual assets is what preserves the cross-sectional correlation structure — on any resampled day, every stock's return is the return it actually had on some real day, so the relationships between them survive. Block length 11.07 sessions was selected by the automatic rule of Politis and White (2004), as corrected by Patton et al. (2009), at bandwidth 14; blocks rather than individual days are what preserve volatility clustering.

Conditioning on regime is what keeps a path plausible: a synthetic history is assembled from blocks drawn within regime, so a calm stretch is built from calm days rather than from an arbitrary mix. 1,000 paths over 1,232 sessions spanning 2020-01-02 to 2024-12-31, at 0.026 seconds per draw.

3.4 Validating the synthetic histories

A resampler that produces statistically valid but economically absurd paths would invalidate everything downstream, so the paths are tested rather than asserted. 9 target moments — volatility clustering, autocorrelation, skew and others — are computed on the real series and on each synthetic one, and the real value must fall plausibly inside the synthetic distribution.

The conditional resampler fails 0 of 9 moments. The unconditional one fails 1, with the real series sitting at the 97.8th percentile of the synthetic distribution on the failing moment. Conditioning on regime is not a refinement — it is the difference between paths that pass their own validation and paths that do not.

We report the unconditional failure rather than deleting the unconditional arm, and the conditioning effect is itself measured: the two resamplers agree at Spearman 0.991 on across-path fragility and 0.976 on across-regime fragility, so conditioning changes the level of the measurement more than it changes the ordering.

3.5 Two tiers, and two definitions of fragility

Fragility can be measured across two different kinds of environment variation, and the benchmark computes both because they answer different questions.

- **Across regimes** — how much does performance move between the 4 labelled market states of the *real* history? Cheap. The charter's primary definition.
- **Across paths** — how much does performance move between synthetic histories that never happened? Expensive: 100 paths, 124.5 CPU-hours, 4617 seconds per path.

These are not two estimates of one quantity. They agree at Spearman 0.550 ($n = 125$), which is low enough that a paper reporting only one of them under the unqualified word "fragility" would be making a choice it had not disclosed. Both are reported throughout, and never pooled.

3.6 Reproducibility

Every number in this paper is generated from committed artefacts by a documented command sequence and substituted into this document by macro. None is transcribed. The full pipeline runs from `data/raw/` and writes a manifest per execution containing the git SHA, package versions, model tags, seeds and input hashes. A cross-check on the warm-up window compared 55 quantities and found 0 beyond tolerance, with a worst relative difference of 4.7×10^{-8}.

4 Which shortcuts are safe

[GENERATION METHOD] **A shortcut reproduced performance. It did not reproduce fragility.**

124.5 CPU-hours is a lot for a measurement that a cheaper procedure might approximate. The obvious shortcut is to skip re-simulating the strategy on each synthetic path and instead resample the strategy's *realised return series* directly — vastly cheaper (23.1 seconds for 1,000 paths, against 4617 seconds for one). Whether that is legitimate is an empirical question, and we answered it by running the expensive version anyway and comparing.

Takeaway. The cheap tier is an excellent proxy for *how well a strategy does* and a poor proxy for *how much that varies*. Since the second is the entire point of the benchmark, the cheap tier cannot replace the expensive one.

Table 1. Cheap tier against full counterfactual re-simulation, $n = 125$.

Quantity	Rank agreement	Median value
Mean performance	0.897	—
Fragility	**0.620**	0.360 vs 0.311

What explains the divergence, and what we will not claim. The two tiers differ in exactly one respect: in the expensive tier the strategy *re-decides* on the path it is given, and in the cheap tier it cannot. The divergence is therefore *consistent with* strategy re-decision contributing

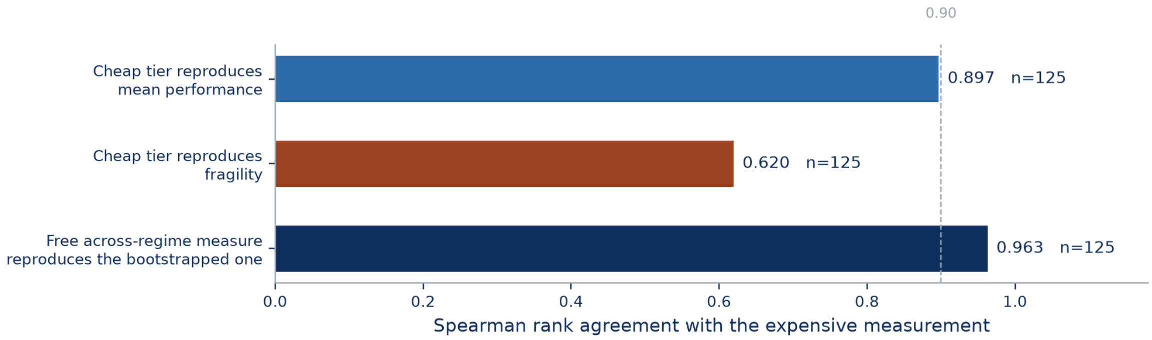

Figure 2. Three shortcuts, measured against the expensive computation they would replace, on the same strategies. Rank agreement, Spearman.

materially to measured fragility — and it is fragility rather than performance that the cheap tier loses, which is what that explanation predicts.

What this does not say

We do not claim causation from a rank correlation. The observation is that two procedures differing in one respect disagree on one quantity and agree on another. That is an empirical methodological result with a practical consequence — *run the expensive tier* — and it is not a demonstration of the mechanism.

4.1 A shortcut that does hold

Not every shortcut fails, and the benchmark distinguishes them by measurement rather than by temperament. Across-regime fragility computed directly from a strategy's persisted return series agrees with its bootstrapped counterpart at Spearman 0.963 ($n = 125$), with medians 0.618 and 0.551.

The expensive experiment was run in full and then used to *validate* the free computation, rather than being replaced by an assumption about it. Convergence in path count confirms the expensive tier is itself converged: rank agreement against the full answer rises from 0.817 at 10 paths to 0.952 at 100 and 0.984 at 500.

Takeaway. One shortcut is validated at 0.963 and adopted; another is measured at 0.620 and rejected. **Which is which was not predictable in advance**, and that is the argument for running both.

5 Three properties of the population

These are facts about the strategies and about the metric. They also appear in Threats to Validity, because each limits what the rest of the paper can claim — and they are stated here as well because a reader who met them only among the caveats would reasonably conclude we had tried to keep them quiet.

5.1 A large minority of machine-written strategies are not deterministic

> [LOCAL 7B] **27 of 185 strategies return a different result when run twice on the same data with the same code**, with a worst-case Sharpe swing of 1.561. Every affected strategy is generator output. **Every standard academic factor is deterministic.**

This is a property of what the model wrote, not of the engine or the harness, and it has a direct consequence for the whole enterprise: **a strategy whose score already varies for no reason is arithmetically indistinguishable from one whose score varies because the market changed.** Nondeterminism and fragility are the same measurement until the first is excluded.

The exclusion is non-random and we say so. The benchmark measures fragility on the subset of machine-written strategies that happen to be reproducible, and reproducibility may correlate with the simplicity that also drives fragility. The census was frozen before any stress path was run.

5.2 The stability filter removes the fast end of the corpus

A strategy is **knife-edge** if its Sharpe ratio is not stable under a numerically negligible change to its inputs — a relative perturbation on the order of 10^{-15}, which no economic interpretation survives. 31 of 156 qualify, including 1 standard academic factor. The rule was applied without exemption.

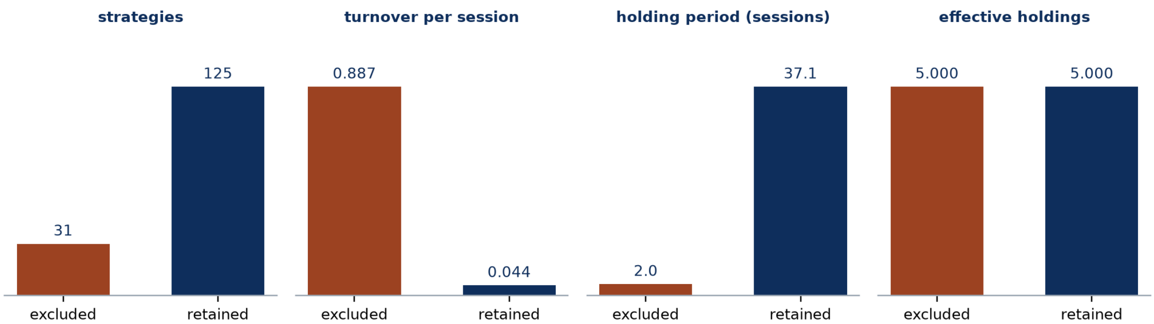

Figure 3. How the knife-edge exclusions differ from the strategies that were retained, on four characteristics with four different units. **The filter did not remove a random slice — it removed the fast end of the corpus**, and the paper's honesty depends on saying so, because the retained population is therefore slower-trading than the one that entered.

> **The exclusion is not random, and that is a limitation of every result downstream**
>
> The excluded strategies are systematically the *fast* end of the corpus: twenty times the turnover and a twentieth of the holding period of those retained. Every result downstream is therefore conditioned on a restricted range of exactly the characteristic most likely to drive fragility. Sharpe swing among the excluded has a median of 0.0088 and a maximum of 2.514, with 4 above 0.5 — so for most of them the instability is small, and for a few it is not.

5.3 A strategy that is uniformly bad is measured as stable

Fragility is a ratio with a mean in its denominator, and it scores consistency, not quality. Nothing prevents a strategy that loses money in every regime from being uniformly consistent

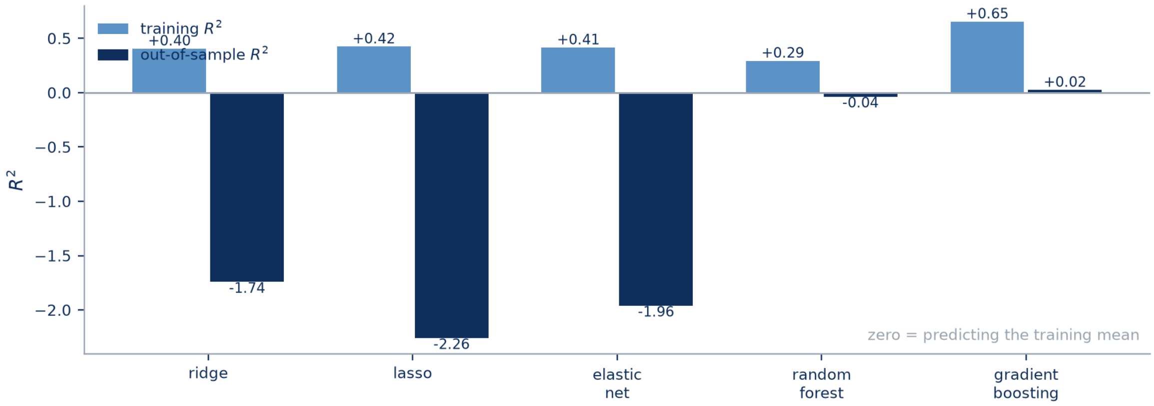

Figure 4. Five model families on identical folds, $n = 109$. Training against out-of-sample R^2 on the primary target. Zero is what predicting the training mean scores.

about it.

> **This is not hypothetical and it is not rare.** 16 of 125 strategies have a negative Sharpe ratio in *every* labelled regime. The most extreme, `candidate_244`, ranges from -2.501 to -1.486 across the 4 regimes — and with a fragility of 0.093 it ranks 5 of 125 on the low-fragility end of the corpus. **It is among the most robust strategies we measured, and it never makes money.**

A fragility score is therefore uninterpretable without a performance figure beside it, and must never be used as a standalone ranking. Our own narrative layer does not enforce this and describes one such strategy as most reliable in a regime in which it loses money (§8). A further 3 strategies are flagged because a near-zero mean in the denominator inflates the ratio for arithmetic reasons; how many sit close enough to that boundary for the flag to miss them is unknown.

6 Predicting fragility, and why it fails

The benchmark's first task: predict a strategy's fragility from its characteristics alone — turnover, holding period, concentration, book-similarity decay, and factor exposures obtained from one joint regression on all 10 factors at once — *without running the stress suite.* If that worked it would be genuinely useful, since the stress suite costs 124.5 CPU-hours and the characteristics are free.

6.1 It does not work

Takeaway. Every model fits its training folds far better than it generalises, and the gap is *widest* for the model with the most capacity. That is the signature of variance rather than bias — which is why we added no capacity, and why a larger corpus rather than a larger model is the correct next experiment.

109 strategies, 26 features, five model families on identical folds.

Figure 4 carries all ten figures. The gap between the paired bars is the diagnosis, and it is why we report the null as a property of the sample rather than of the model family.

Takeaway. The best out-of-sample R^2 is $+0.024$, from gradient boosting (Breiman, 2001). Predicting the training mean would score zero. This is not a model that needs tuning; it is a problem that is not solvable at this sample size.

6.2 The diagnosis: variance, not bias

Reporting a null without diagnosing it is not useful, because the reader cannot tell whether to try harder or to stop. We separate the candidate explanations by measurement.

Table 2. Candidate explanations for the null, each measured.

Explanation	Measurement	Verdict
Heavy-tailed target	Skew 5.72, kurtosis 32.1; top 5% of observations hold 96.1% of the variance	Real. A $\log(1+x)$ transform lifts the best R^2 to +0.092 — still weak.
Multicollinear features	Condition number 253.2; worst pair `mean_herfindahl` and `mean_largest_weight_share` at $r = 0.995$	Present, and it degrades the linear models specifically.
Influential observations	Dropping the 5 most influential moves R^2 by −35.864; the 10 most, by −124.861 (Cook, 1977)	Severe. The result is not stable to a handful of rows.
Too few strategies	Learning curve: +0.138 at 40 rows, +0.247 at 100	**Still climbing at the full sample.** More data would help; more capacity would not.
No relationship at all	Permutation test, 200 shuffles (Ojala and Garriga, 2010): $p = 0.020$ on rank, $p = 0.861$ on level	**Rejected for ranking, not rejected for level.**

Strategy characteristics contain statistically detectable *ranking* information and very weak information about the *level* of fragility. The permutation test rejects the null for rank order ($p = 0.005$–0.020) while failing to reject it for the level on three of four targets.

Every model fits its training folds far better than it generalises, and the gap is widest for the model with the most capacity — gradient boosting at 0.629 against the random forest's 0.329. That is the signature of variance rather than bias. **We therefore added no capacity**, which is the action the diagnosis implies and the opposite of what a search for a better number would have produced.

The five characteristics that carry the most weight, reported for completeness rather than as an economic finding: `beta_long_term_reversal_756d` (+0.408), `factor_r_squared` (+0.280), `trading_session_rate` (+0.187), `mean_holding_period` (+0.176) and `beta_inverse_volatility_` (+0.148).

7 Two corrections, reported as findings

Our first answer was better, and wrong

The first version of the fragility predictor produced a modest positive result. It was wrong, and we found out by deduplicating the corpus.
Correction 1 — duplicate leakage. 11 clusters of strategies produce *identical* realised return series, covering 27 of 150 compared strategies, with a largest cluster of 4. A duplicate sitting in a training fold with its twin in the test fold is not generalisation — it is memorisation with the answer key in the room. Deduplicating (16 rows removed, 6 not comparable) moved the Spearman result from +0.262 at $n = 125$ to +0.024 at $n = 109$. **The duplicates were worth 0.238 of R^2**, which is to say: the entire original result.
Correction 2 — a duplicated feature. One feature column duplicated another under a different name, inflating that characteristic's apparent importance.
Both corrections made every reported number worse. Both are in the benchmark's corrections log. And the duplicate structure ships as a benchmark artefact rather than being quietly removed, because **it is a fact about machine-written strategy corpora, not a blemish on ours** — a near-duplicate rule at threshold 0.9999 fires a further 20 times below the exact criterion.

8 Stress narratives

A local model (qwen2.5:7b-instruct-q4_K_M) generates natural-language descriptions of when a strategy fails — the explainability layer that would make a fragility number presentable to a risk committee. 12 narratives at 5.5 seconds each.

What this does not say

The narrative layer is the least validated component in this paper and we do not recommend relying on it. It reads a fragility score and a set of regime performances and produces fluent prose about them; nothing checks that the prose is warranted. It described one of the 16 uniformly-losing strategies of §5.3 as most reliable in a regime in which it loses money. That sentence is true about consistency and misleading about everything a reader would take from it.

9 Does any of this depend on who wrote the strategies?

Everything above was measured on a corpus written by one local seven-billion-parameter model. The objection is obvious and it is the right one to raise: this paper may be describing that model rather than machine-written strategies in general.

So the identical frozen task specification was issued to 3 frontier models (4 independent requests each) and all 60 strategies were passed through this benchmark **with no parameter altered and no line of measurement code added**. The same benchmark was separately applied to 6 generation methods on the local model.

Takeaway. Fragility does not move much with the generator, and does not move much with the generation method either. It is closer to a property of the market than of the author.

9.1 Fragility is generator-independent

[GENERATOR-INDEPENDENT] **Fragility is a property of the market, not of the author.** The primary measure is 0.618 on the local corpus against 0.566, 0.624 and 0.696 on the three frontier arms. **Every arm lands within a sixth of the local median.** This is the result the benchmark most needed and did not have: what it measures survives a change of generator.

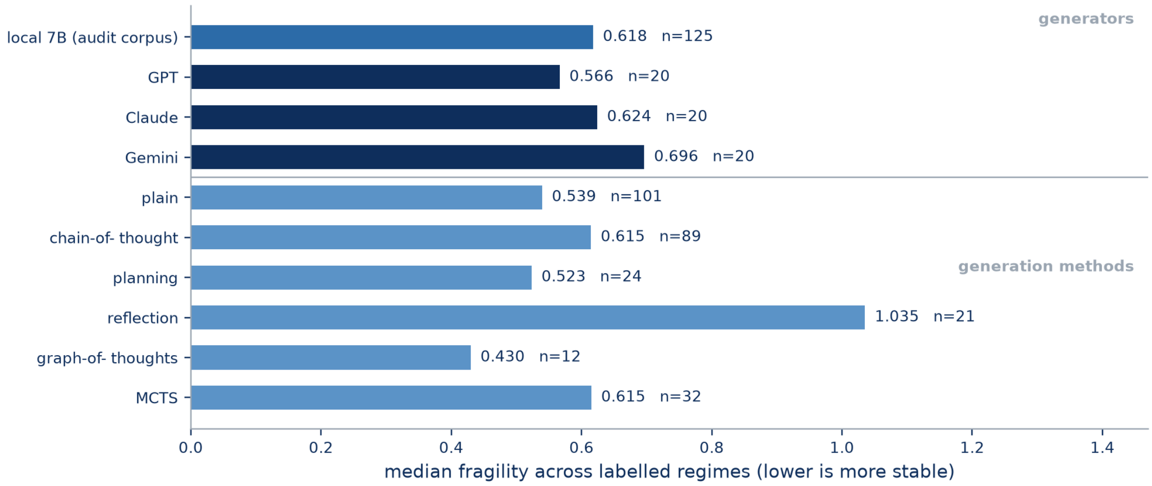

Figure 5. Median fragility across labelled regimes — one definition, one axis. The upper block varies *who* wrote the strategy; the lower block varies *how* the same model was prompted. Sample sizes are printed on every bar because they vary by an order of magnitude.

9.2 The knife edge recurs; nondeterminism does not

Between fifteen and thirty-five per cent of each frontier arm sits on a knife edge, against 31 of 156 locally. The worst frontier case moves by more than two Sharpe units under a fifteenth-decimal-place perturbation. §5.2 argued this is a property of how strategies are written rather than of any one writer, and the frontier arms support that.

> [FRONTIER] **Nondeterminism does not recur.** Not one of the 60 frontier strategies changes its result across two hash seeds or across a replicate at a fixed seed, against 27 of 156 locally. Iterating an unordered container is a habit the frontier models do not have. **It is the one measured pathology in this paper that is genuinely a property of the generator**, and it is the reason §5.1's exclusion is a statement about the local model rather than about machine-written code.

9.3 The predictor's failure is not local either

§6 reported that fragility cannot be predicted from strategy characteristics. Trained unchanged on the same 109 local strategies and applied to the 60 frontier strategies as held-out data, the model reaches a pooled out-of-sample R^2 of -0.004 against its training mean: **no better than a constant, on data from a different generator.** The negative result is a property of the problem.

What this does not say

One caveat we did not expect, reported as **exploratory** since it appears in no pre-registration: while the level predictions are worthless, the rank ordering is not. Spearman correlation between predicted and measured fragility is +0.717, +0.540 and +0.556 on the three arms and +0.555 pooled. The model cannot say how fragile a strategy is; it partially orders which of two is more fragile. **At 20 strategies per arm this is a direction worth testing properly, not a finding.**

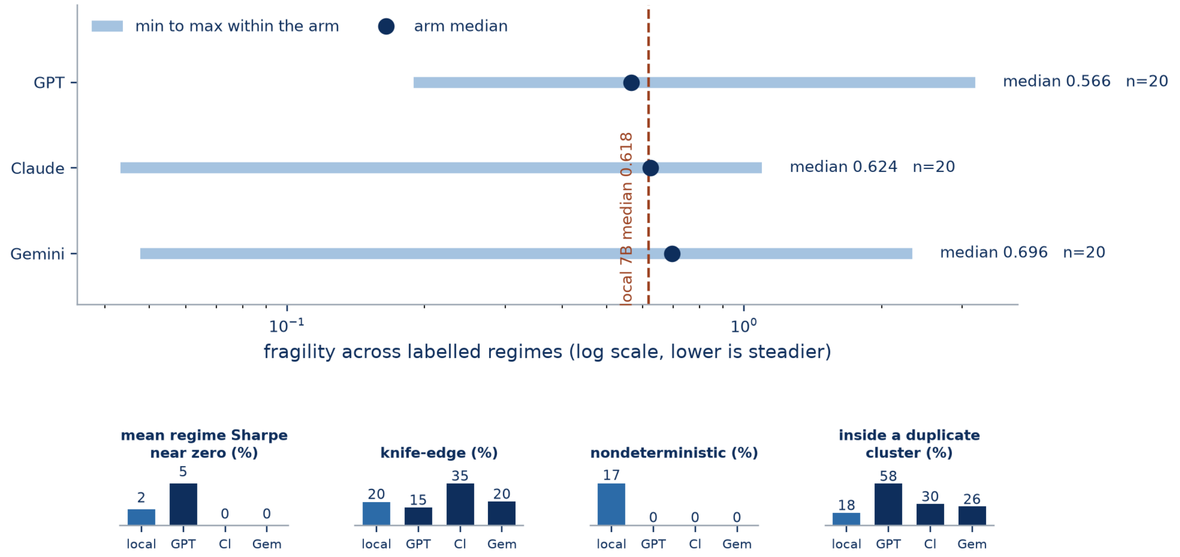

Figure 6. Every RegimeStress measurement, by generator. *Upper:* fragility across labelled regimes — the charter's primary definition — with the full range each median summarises, on a log axis. **The spread inside any one arm is far wider than the gap between arms.** *Lower:* the three pathologies and the duplicate structure, as rates so that a 20-strategy arm and the 156-strategy local corpus sit on one scale. `Cl` is Claude and `Gem` is Gemini.

9.4 Duplicate clustering persists everywhere

11 of 19 comparable GPT strategies sit inside an exact-duplicate cluster, 6 of 20 on Claude and 5 of 19 on Gemini — across requests issued in independent sessions. The near-duplicate rule that the exact criterion does not merge fires 3 times on Claude and 0 and 0 times on the other two.

The redundancy finding of §7 holds at the frontier. Paper 1 reports the sharper version of this: strategies from two *different vendors* whose realised return series are identical to floating-point precision.

9.5 Does the generation *method* change fragility?

The same benchmark, applied to 6 generation methods on the local model, all using the across-regime definition.

The lower block of Figure 5 carries the six arms on the same axis and the same definition as the upper block, so the two questions — does the generator matter, does the method matter — can be read against one another without rescaling.

Takeaway. Reflection is the outlier at 1.035 on 21 strategies — the only method whose median exceeds one, meaning its strategies' performance varies by more than its own average size. Every other method sits in a band from 0.430 to 0.615. We report the reflection figure without an explanation for it.

What this does not say

These are not comparable to the tier-1 across-path figures, which on the same frontier arms are 0.418 for GPT (range 0.189–32.785), 0.355 for Claude (0.169–1.373) and 0.334 for Gemini (0.164–2.703). The two definitions must never be plotted on one axis. §3.5 measured their agreement at 0.550 ($n = 125$), which is too low for one to stand in for the other. Every arm in both blocks of Figure 5 is an across-regime figure, matching the upper panel of Figure 6 and nothing else in this paper.

Nor is the local reference in the two blocks the same population. Figure 6 uses paper 1's pooled audit corpus ($n = 125$); the plain-prompting arm in Figure 5's lower block is a separate 830-draw arm generated for the methodology experiment ($n = 101$). They are two samples from the same generator under the same prompt, and their medians differ.

10 The standard-factor control, and a reversal

[EXPLORATORY] **The published factors are *more* fragile than the machine-written corpus** — median 0.8581 across regimes against 0.618 ($n = 11$ against $n = 125$). The metric runs backwards from the direction anyone would predict, and the reason is §5.3.

§9 varied the generator. This section varies something else: the population includes 11 standard factor strategies — momentum, low volatility, short-term and long-term reversal, relative strength, an equal-weight market reference, and two deliberate controls. They are published, long-standing, and were never drawn from any search of ours.

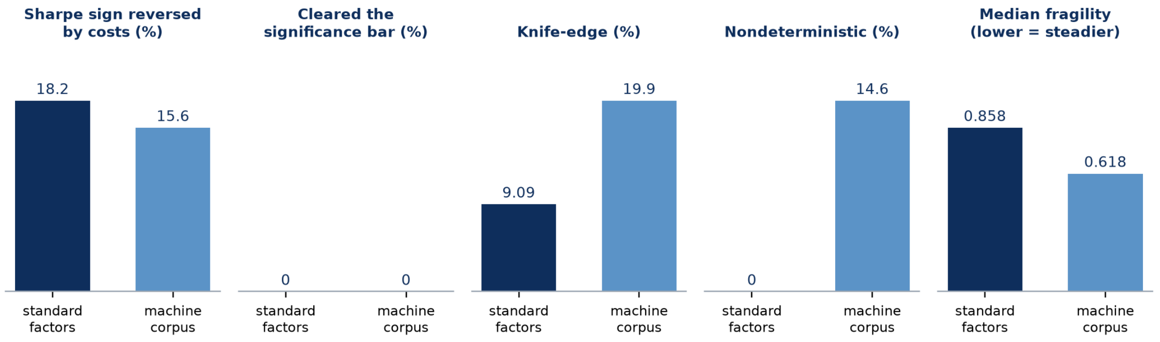

Figure 7. The standard-factor arm beside the machine-written corpus on five frozen tests, each panel on its own scale because the five quantities share no unit. Sample sizes differ by an order of magnitude — 11 against 156 — and the two reversals, costs and fragility running opposite ways, are the reason this is drawn rather than tabulated.

Why fragility runs backwards, and why that is the metric's fault rather than the corpus's. A published factor takes real positions and its performance genuinely differs between a crisis and a calm stretch — which is what fragility is supposed to measure. Much of the machine corpus is near-flat or uniformly mediocre, and a strategy that does approximately nothing does approximately the same nothing in every regime. It therefore scores as admirably stable. §5.3 showed 16 of 125 machine strategies losing money in *every* regime, one of them ranking 5 of 125 for robustness.

The standard-factor arm converts that from an argument into a measurement. A population known in advance to contain real strategies scores *worse* on our fragility metric than a population known to be mostly inert. **Any use of fragility as a standalone ranking would have preferred the inert one.** The metric is not wrong; it answers a question about consistency and we must stop reading it as a question about quality.

Nondeterminism is the one clean separation. 0 of 11 standard factors are nondeterministic, against 27 of 156 machine-written strategies and 0 of 60 frontier ones. Combined with §9, the picture is consistent: nondeterminism is a habit of one small model, absent from published code and absent from frontier output.

What this does not say

Eleven strategies. Every rate here has a denominator of eleven, so one strategy moves it by nine points. The arm is a control, not a sample. **It is also not a "human versus AI" comparison**: the contrast is *published and long-standing* against *machine-written*, and these fixtures were hand-written for a different purpose by an author who knew the evaluation. The deflation column and this assembly appear in no pre-registration and are exploratory.

11 Design flaws, mistakes, and things that did not work

Table 3. Defects in our own apparatus and results that failed, with what each cost.

Defect or null	What it cost	Where
Duplicate leakage	11 clusters of identical return series split across training and test folds were worth 0.238 of R^2 — the whole of our original positive result.	§7
A duplicated feature column	One characteristic appeared twice under different names, inflating its apparent importance.	§7
Fragility scores consistency, not quality	A strategy losing money in every regime ranks 5 of 125 for robustness. Now measured against a control arm that scores *worse* than the inert corpus.	§5.3, §10
Both exclusions are non-random	Knife-edge removes the fast end of the corpus (turnover 0.887 against 0.044); nondeterminism removes a subset that may differ systematically. Every result is conditioned on a restricted population.	§5
The narrative layer is unvalidated	It described a money-losing strategy as most reliable in a regime where it loses money. We do not recommend relying on it.	§8
Regime labels drift	21.8% disagree with the terminal fit. The honest cost of refusing hindsight, and it means the partition itself is uncertain.	§3.2
Synthetic histories are an assumption	Every across-path number inherits it. 0 moment failures of 9 is the strongest evidence we can offer and is not the same as being right.	§12

11.1 Results that failed

- **Fragility cannot be predicted from strategy characteristics.** R^2 =+0.024 at $n = 109$, and -0.004 on frontier strategies as held-out data. This was the benchmark's first task.
- **The cheap stress tier cannot replace the expensive one** for the quantity of interest: 0.620 on fragility against 0.897 on performance.
- **Our first predictor result was better and wrong**, by 0.238 of R^2.
- **Fragility does not separate generators** — which we report as a success for the benchmark and a null for anyone hoping to rank models by it.
- **Our own metric prefers strategies that do nothing**, demonstrated in §10 on a control population.

12 Threats to validity

These are the reasons a reader should discount what is above. They are a section rather than a footnote because several are load-bearing.

T1 — The synthetic histories are an assumption, not evidence. Across-path fragility rests entirely on the resampler being a reasonable model of how markets could have gone. It passes 0 failures on 9 moments, which is the strongest evidence we can offer and is not the same as being right. **This is the paper's single largest assumption** and every across-path number inherits it.

T2 — Both exclusions are non-random. Knife-edge exclusion removes the fast end of the corpus (§5.2); nondeterminism exclusion removes strategies that may differ systematically from those retained (§5.1). Every downstream result is conditioned on a restricted population.

T3 — The regime labels drift. 21.8% of causal labels disagree with the terminal fit. That is the honest cost of not using the future, and it means the regime partition a strategy is scored against is itself uncertain.

T4 — Sample size bounds the prediction task, and we know it does. The learning curve is still climbing at 100 rows. A larger corpus is the correct next experiment, and it is not one we can run under this programme's constraints.

T5 — Fragility scores consistency, not quality. §5.3. Any use of these numbers as a standalone ranking is a misuse, including by our own narrative layer.

T6 — One market, one window, one corpus. 2020-01-02 to 2024-12-31, Indian large-cap equities, a corpus of machine-written strategies. Factor exposures are computed against 10 factors with a design condition number of 12.0 and a worst pair correlation of 0.946, so the exposures are not cleanly separated from one another.

T7 — The frontier arms are 20 strategies each. Every claim in §9 rests on 60 strategies in total. They are enough to show that fragility does not move dramatically with the generator; they are not enough to establish that it does not move at all.

13 What this tells us about AI-driven research

Three claims, at three scopes, kept separate.

Generator-independent [GENERATOR-INDEPENDENT] Fragility itself, the fragility-prediction null, and the knife-edge pathology all reproduce across four generators. These are the claims that license the benchmark being used by other people.

Generator-specific [LOCAL 7B] Nondeterminism is a habit of the local model and of no frontier model. **An exclusion we described as a property of machine-written strategies is in fact a property of one model**, and §9 is where we correct our own framing.

Method-independent [GENERATION METHOD] Fragility does not move much with the generation method either, on samples of 12 to 101. Making the AI think harder did not make its strategies more robust.

> Paper 1 found that a better generator writes better code without producing better evidence. Paper 2 adds the complementary result: **a better generator does not write more robust strategies either.** Capability moved executability from 14.6% to 100% and moved median fragility by less than a sixth.

14 The RegimeStress benchmark

Frozen and released at `github.com/Ashwin-aadi/Trivijaya-Quant-Research`:

- **The regime series.** 1,233 causally-labelled sessions, 20 expanding-window refits, plus 18 dated and sourced structural events.
- **The resampler.** CRR with its block-length rule, its conditioning, and the 9-moment validation suite it must pass.
- **The population.** 125 strategies with fragility on both definitions, plus the 31 knife-edge and 27 nondeterministic exclusions shipped rather than discarded — they are the benchmark's hardest cases.
- **The duplicate structure** as an artefact, since it is a fact about generated corpora.
- **Every number**, regenerable from `data/raw/` by documented commands. If this paper and the repository disagree, the repository is right.

15 Conclusion

A strategy that survived 2020–2024 survived one sequence of events. We built the apparatus to ask what would have happened under sequences that did not occur, validated the synthetic histories against the real one, and measured fragility on a frozen population with its exclusions fixed in advance.

The methodological result is that **a shortcut reproduced performance and did not reproduce fragility** — Spearman 0.897 against 0.620 on the same 125 strategies — so the expensive tier is not optional, and we know that because we ran it. The prediction task is a diagnosed null: R^2 =+0.024, bounded by sample variance rather than model bias, and reproducing at -0.004 on strategies from three other generators. And fragility proved to be a property of the market rather than the author, which is the property a benchmark needs before anyone else can rely on it.

> **Our own first answer was better and wrong**, by 0.238 of R^2, because 11 clusters of identical strategies were split across training and test folds. We report that in the results. A benchmark that hides its own corrections is not measuring anything.

Where this leaves us

Papers 1 and 2 asked whether a strategy is real and when it breaks. Both questions are asked at a fixed, small book size — Rs. 10 lakh — and the answers are conditional on it in a way that is easy to miss. Paper 1 already found one standard factor whose verdict *reverses* between Rs. 10 lakh and Rs. 10 crore, because flat per-scrip fees dominate small books and market impact dominates large ones.

So a strategy that is real and robust is still not deployable until one more question is answered, and it is the question a portfolio manager asks first: **how much money can it actually absorb?** Run Rs. 10 crore through a strategy that works at Rs. 10 lakh and your own trading moves prices against you until the edge is gone. Nobody has published those numbers openly for Indian markets.

That is the third paper, **FlowState** — which also reports, as a principal finding, why daily data cannot answer the version of that question everyone asks first.

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